Literature Review: Set Theoretic Estimation

Working notes toward a Lean mechanization

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Abstract

Working notes reviewing set theoretic estimation (STE) from its unifying statement [4] through its topological-space, cryptographic instantiation [3], oriented toward identifying which results are ripe for machine-checked formalization in Lean 4 / Mathlib. These notes are exploratory and cite the primary sources; only claims discharged in the CI-checked `Ste` library are asserted as proven.

1 The set theoretic paradigm (Combettes 1993)

Conventional estimation optimizes an objective function; the solvability and realism of that optimization is a persistent difficulty [4]. STE replaces optimization with *feasibility*.

Fix a *solution space* Ξ and a true object (estimandum) h. Each piece of information $i \in I$ is a proposition on Ξ; at a chosen confidence level it yields a *property set*

$$S_i = \{a \in \Xi \mid \Psi_i(a) \geq \psi_i\}, \qquad \Psi_i : \Xi \to [0,1], \tag{1}$$

the set of estimates consistent with information i (Combettes Eq. 3). The *feasibility set* is their intersection,

$$S = \bigcap_{i \in I} S_i \qquad \text{(Combettes Eq. 4).} \tag{2}$$

Any $a \in S$ is a *set theoretic estimate*. The formulation is *consistent* if $S \neq \emptyset$, *fair* if $h \in S$, and *ideal* if $S = \{h\}$. When Ξ is a real Hilbert space and each S_i is closed and convex, S is closed convex and can be reached by projection algorithms (POCS, 6; cyclic projections, 2; survey 1).

What we formalized. The *crisp* core — where Ψ_i is $\{0,1\}$-valued so S_i is an ordinary subset — is machine-checked in `Ste.Basic`: feasibility set, membership, refinement of each constraint, information monotonicity, the fair/consistent link, the inconsistency⇒invalid-information self-diagnostic, and the ideal-information identification theorem. Convexity of the feasibility set is `Ste.Convex`. The graded/fuzzy case and the projection *algorithms* (as opposed to the geometry) remain open.

2 Topological-space STE for decryption (Carlson 2012)

Carlson [3] makes two moves. First, a *spatial* move: prior STE works in a Hilbert / metric space and simplifies processing with Optimal Bounding Ellipsoid (OBE) algorithms, but an OBE is not confined to the original feasibility volume and can spuriously *expand* the solution set. Carlson instead works in a *topological space*, where property sets are static a priori information that never expand, avoiding the OBE pathology. Second, an *application* move: STE is applied to decryption for the first time.

The decryption reduction. The solution space is a finite key space K. Each intercepted ciphertext c induces a property set of keys under which c decrypts to a meaningful message; the feasible key set is the STE feasibility set of these property sets, and decryption succeeds when it collapses to the true key — the set theoretic form of Shannon's unicity [5].

Named results (extracted from the dissertation text).

- **Lemma 2.1:** the error e (number of keys lacking an encryption property Φ_i) is bounded, $0 \leq e \leq |K| - 1$.
- **Lemma 2.2:** property sets behave as independent variables whose intersection is no larger than the smallest set; if $\Phi_j \subseteq \Phi_k$ then Φ_k is subsumed.
- **Theorem 2.3:** each property set is an AEP typical set; hence information theory falls under the STE umbrella.
- **Lemma 4.1:** a substitution (S) cipher on message M admits $(|A| - |T|)!$ isomorphic (equivalent) keys, where T is the set of distinct symbols occurring in M.
- **Theorem 4.1:** a permutation (P) cipher admits $\binom{|S_t|}{C}(|B| - |S_t|)!$ unique keys, S_t the static-bit set.
- **Theorem 5.2:** a permutation encryption reduces to a substitution encryption within a symbol's bit representation.
- **Corollary 5.3:** a boundary-aligned PSP cipher is idempotent to a single S cipher; therefore every block cipher is a block substitution cipher, so one attack (BCBB) suffices.

What we formalized. The topological substrate — closedness and, via compactness and the finite intersection property, existence of feasible points — is `Ste.Topology`; it specializes cleanly to Carlson's finite (discrete, hence compact) key spaces. The STE-decryption skeleton (cipher systems, key property sets, feasible keys, fairness of genuine traffic, unicity as ideal information, observation monotonicity) is `Ste.Carlson.Cipher` (namespace `STE.Carlson`). Beyond the skeleton, two of Carlson's named results are now *proven* in the CI-checked library:

- **Lemma 4.1** (residual key ambiguity): `STE.Carlson.card_consistent_keys` shows the keys of a substitution cipher agreeing with the true key on the observed symbol set $T \subseteq A$ number $(|A| - |T|)!$, i.e. the size of the STE feasible key set after one observation (with `card_substitution_keys`: $\#\mathrm{Perm}\, A = |A|!$). File `Ste.Carlson.Counting`.
- **The reduction chain (Theorem 5.2 / Lemma 5.1 / Corollary 5.3):** `permutation_reduces_to_substit` and `coperm_injective` embed permutation-cipher keys injectively into substitution-cipher keys; `card_permutation_lt_card_substitution` and `substitution_not_reducible_to_permutation` show $n! < (2^n)!$ for $n \geq 2$, so the embedding is non-surjective (a substitution cipher need not be a permutation cipher); and `psp_reduces_to_substitution` shows every PSP composition is a single substitution cipher. File `Ste.Carlson.Reduction`.

3 Assessment: what is ripe to mechanize

1. **Done (this revision):** the cipher counting Lemma 4.1 and the reduction chain (Theorem 5.2 / Lemma 5.1 / Corollary 5.3), both finite combinatorics/algebra over `Fin`-indexed symbol/bit alphabets built on Mathlib's `Fintype.card_perm` ($|\mathrm{Perm}\,\alpha| = (\#\alpha)!$) and factorial API. Remaining in this family: Carlson's general Theorem 4.1 (the $\binom{|S_t|}{C}(|B| - |S_t|)!$ count).

2. **Analytic:** POCS convergence (1, 6) atop `Ste.Convex`, using Mathlib's Hilbert-space projection theory.

3. **Deepest:** Theorem 2.3 (property sets as AEP typical sets), which requires a probabilistic model and Mathlib's measure/information theory.

4 Distrust discipline

Per project rule, no result here is asserted proven unless it compiles in the `Ste` library checked by CI on every push to `main`. Carlson's Lemma 4.1 (H9) and the reduction chain (H10) are now machine checked; the remaining named results (Theorem 4.1, Theorem 2.3) are *quoted from the source* as targets. Statuses live in `log/2026-07-19-hypotheses-ste-core.md`.

References

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